

Aspirin in Audiology: A Dual-Edged Sword in Hearing Loss and Its Nanotechnology-Driven Future

Yang Yang, Chaoyong Tian, Xiaogang An, Bei Fan, Dingjun Zha

Department of Otolaryngology-Head and Neck Surgery, Xijing Hospital, Air Force Medical University, Xi'an, 710032, People's Republic of China

Correspondence: Dingjun Zha, Email zhadun@fmmu.edu.cn

Abstract: Globally, more than 1.6 billion individuals are affected by hearing loss. Aspirin (acetylsalicylic acid, ASA), an inexpensive and widely available drug, has demonstrated a complex dual role in hearing loss, with its effects potentially influenced by dosage and individual variability. In this article, we aimed to provide an overview of the pharmacological properties of aspirin, followed by an in-depth discussion of its mechanisms of action and its toxic and protective effects in different types of pathology-induced hearing loss. Finally, the development of novel drug delivery systems that may enhance the use of this drug in preventing hearing loss was also discussed. Aspirin exerts both ototoxic and protective effects via cyclooxygenase (COX)-dependent and COX-independent signaling pathways, including Wntless/Integrated (Wnt) signaling, nuclear factor kappa-B (NF- κ B), and Prestin-related mechanisms. Low-dose aspirin appears to reduce hearing damage from noise exposure or ototoxic drugs through anti-inflammatory and antioxidant actions that limit cochlear oxidative stress and inflammation. Conversely, higher doses of aspirin may induce reversible auditory changes, such as temporary hearing threshold shifts, tinnitus, and synaptic damage. These effects are primarily associated with cochlear ischemia, excessive activation of N-methyl-D-aspartate (NMDA) receptors, impaired prostaglandin signaling, and altered outer hair cell function resulting from Prestin modulation. This dose-dependent paradox presents a major challenge for the clinical application of aspirin in hearing protection. Nanotechnology-based delivery systems and personalized dosing strategies are promising, although they remain largely at the preclinical stage. Future work should concentrate on optimizing doses and targeting drug delivery to cochlear blood vessels or outer hair cells. Particular attention should be given to aspirin's role in COX signaling pathways and Prestin structural regulation. Overall, these findings provide a theoretical basis for precision-based ear protection strategies, pending clinical validation.

Keywords: aspirin, hearing loss, nanotechnology, prestin, spiral ganglion neurons

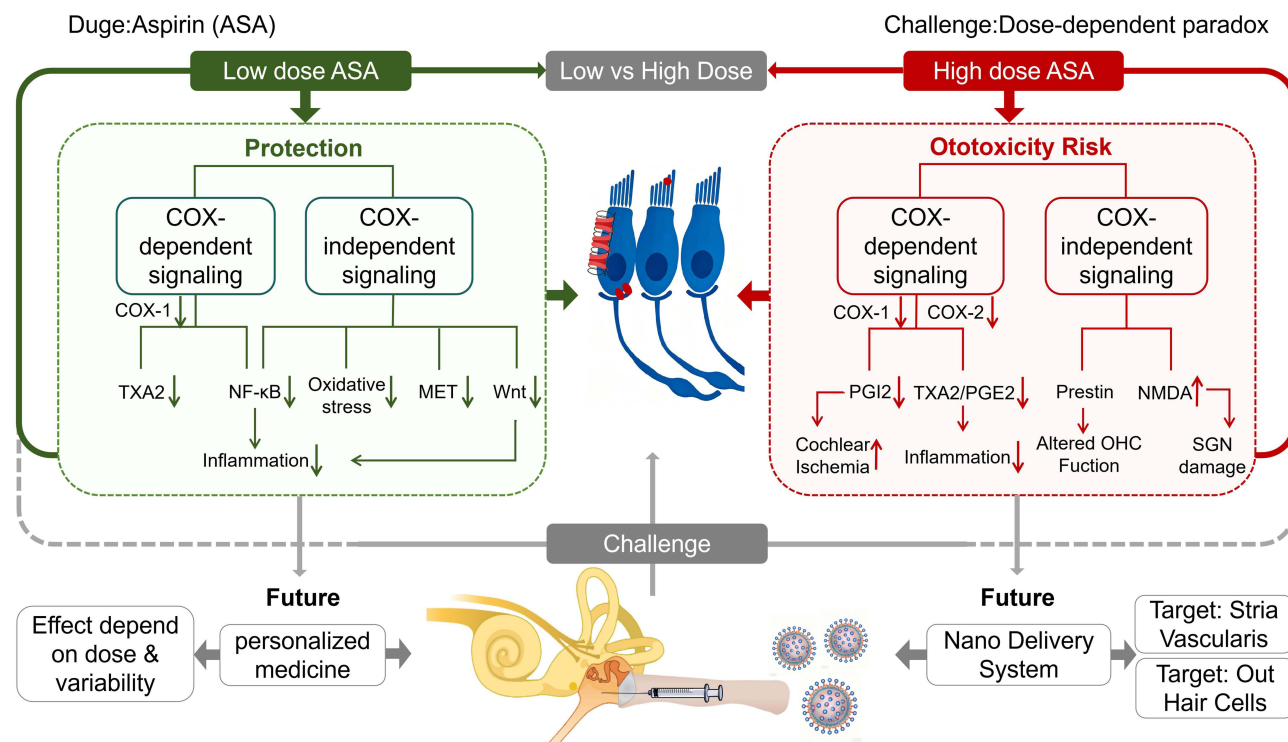
Introduction

Hearing loss represents a significant public health burden, impacting nearly 432 million adults and 34 million children worldwide, or about 5% of the global population.¹ Overall, around 1.6 billion individuals are affected, with more than half of the cases attributable to preventable causes.² Beyond its health impact, hearing loss imposes significant social and economic burdens.³

Aspirin (acetylsalicylic acid, ASA) is a salicylic acid-based drug with a therapeutic history, dating to the late 18th century when willow-derived salicylic acid was commonly employed for pain and fever relief.⁴ Its low cost and global accessibility have contributed to its widespread use in various formulations.⁵

Aspirin is not only an effective analgesic and anti-inflammatory agent but also plays a crucial role in preventing cardiovascular diseases such as arteriosclerosis, heart attacks, and strokes by inhibiting platelet activation and aggregation. Mechanistically, aspirin prevents thrombosis via the inhibition of cyclooxygenase (COX), which reduces thromboxane A₂ (TXA₂) production. In recent years, aspirin has also gained attention as a preventive agent for inflammation-related cancers.⁶ Long-term or low-dose administration of aspirin was found to reduce the risk of colorectal,^{7,8} gastric,⁹ pancreatic,¹⁰ and ovarian cancers.¹¹ Additionally, aspirin has been shown to improve

Graphical Abstract



outcomes in patients with sepsis and related complications.¹² However, high-dose or long-term use of this drug can also cause adverse effects, including tinnitus, gastrointestinal irritation and bleeding, renal dysfunction, liver injury, and allergic reactions.¹³

Given its broad pharmacological effects and known ototoxic potential at high doses, aspirin has drawn increasing interest in audiology. While low-dose aspirin may offer protective anti-inflammatory and antioxidant effects,^{14–17} high doses may contribute to hearing loss.¹⁸ Aspirin and its active metabolite, salicylate, are potent antioxidants. Reports suggest that aspirin may also have neuroprotective and otoprotective effects. Twenty years ago, Sha et al published a randomized trial involving 195 patients with aminoglycoside-induced hearing loss treated with aspirin.¹⁹ Subsequently, our group confirmed that moderate doses of aspirin (975 ± 212 mg) markedly reduce the risk of gentamicin-induced hearing loss.¹⁷ Conversely, inappropriate dosing can cause temporary hearing loss and tinnitus. Conflicting evidence remains regarding aspirin's effects on hearing. While low-dose aspirin shows protective effects in animal models of age-related hearing loss (ARHL), these benefits have not been consistently observed in humans.²⁰ Similarly, its protective potential against cisplatin-induced ototoxicity, demonstrated in preclinical studies, has yet to be confirmed in clinical trials.²¹ Additionally, studies on its effects in noise-induced hearing loss (NIHL) and sudden sensorineural hearing loss (SSNHL) in patients with underlying conditions have yielded inconsistent results. Moreover, the mechanisms behind aspirin's dual effects on hearing, especially its impact on Prestin structure and the balance between hair cell preservation and neural injury, are not yet fully understood. These uncertainties contribute to ongoing debate about its effectiveness and safety in the prevention and treatment of hearing loss.

The objectives of this review are threefold: (i) to critically evaluate the current evidence on the dose-dependent dual effects of aspirin on hearing; (ii) to summarize the molecular mechanisms underlying aspirin-induced otoprotection and ototoxicity, with particular emphasis on COX-dependent and COX-independent pathways; and (iii) to discuss emerging nanotechnology-based delivery strategies that may enhance the therapeutic window of aspirin for inner ear applications, based on recent advances in the field. Based on these insights, we propose a comprehensive strategy for aspirin

nanodelivery integrating four key elements: ROS-responsive release, LS19-mediated OHC targeting, sustained-release formulations, and combination therapy for hearing loss treatment, though this approach requires further preclinical validation.

Pharmacokinetics

Aspirin is a frequently utilized nonsteroidal anti-inflammatory drug (NSAID). The pharmacokinetics of aspirin are relatively simple, as it does not require metabolic activation. Due to its weak acidity, aspirin undergoes rapid absorption in the stomach and upper intestine after oral administration, without the need for active transporters. It then undergoes first-pass metabolism in the liver, where a portion of acetylsalicylic acid is deacetylated to form salicylic acid. Approximately 60% to 70% of the drug is quickly absorbed in the gastrointestinal tract and completely metabolized into the active ingredient salicylic acid. Aspirin has a short half-life of about 15–20 minutes.^{5,22,23} However, salicylic acid exhibits a relatively long half-life of 2 to 6 hours and is predominantly metabolized in the liver. Only approximately 10% of the administered dose is excreted unchanged in the urine. Metabolism produces five main metabolites in varying proportions: two glucuronide conjugates (salicylphenol glucuronide and salicylacyl glucuronide), which together constitute about 15%; two dihydroxybenzoic acids (2,3-dihydroxybenzoic acid and 2,5-dihydroxybenzoic acid), accounting for 2–3%; and salicyluric acid, which represents the largest urinary metabolite, comprising roughly 70% of the excreted dose.^{5,22–24}

Aspirin is available in several formulation forms, with the most used being immediate-release and enteric-coated preparations. The immediate-release formulation provides rapid absorption and bioavailability comparable to the powder form. In contrast, enteric-coated tablets withstand dissolution in gastric acid, delaying absorption until reaching the more neutral pH of the intestine. As a result, the enteric-coated tablets can extend the time required to achieve peak plasma concentrations and maximal antiplatelet effect for about 4 to 5 hours.²⁵ Other less common formulations include effervescent tablets, chewable tablets, and combination products, each offering specific advantages in terms of onset of action, patient tolerability, and dosing convenience.^{26,27}

Research has shown that after intraperitoneal injection of 450 mg/kg sodium salicylate in chinchillas, there is a strong linear relationship between serum and extracochlear perilymph concentrations.¹⁸ This relationship can be described by the equation $P = 0.403(S) - 0.30$ (S: serum concentration; P: perilymph concentration) and shows that in guinea pigs, the perilymph concentration is roughly 40% of the serum concentration.²⁸ The blood-labyrinth barrier (BLB) limits salicylate penetration into the inner ear,²⁹ meaning that high systemic doses are required to reach therapeutic cochlear levels, which in turn increases the risk of toxicity to neural structures, such as spiral ganglion neurons (SGNs) and auditory nerves. Animal body weight also affects serum salicylate levels, likely due to differences in body fat content that influence the drug's distribution volume. As a result, systemic administration cannot reliably control inner ear drug concentrations, and blood salicylate levels vary significantly between individuals. These pharmacokinetic challenges highlight the need for targeted local delivery strategies and personalized dosing approaches. Such strategies could overcome the BLB, achieve effective cochlear drug concentrations, and minimize systemic exposure and toxicity.

Mechanisms of Action

Aspirin affects hearing through four primary pathways: COX-dependent, non-COX-dependent, antioxidation-mediated, and prestin-mediated (Figure 1).

COX-Dependent Mechanism

Aspirin irreversibly inhibits the iron-containing COX enzymes, also known as prostaglandin oxidase reductase, via the acetylation of specific serine residues (Ser530) at the active site. These enzymes are involved in the biosynthesis of prostaglandins and thromboxanes from arachidonic acid (AA).³⁰ COX has two isozymes, COX-1 and COX-2. COX-1 is constitutively expressed in most tissues and plays a key role in regulating vasomotion, platelet aggregation, gastric mucosal blood flow, gastric mucus secretion, and renal function.³¹ COX-1 mRNA is highly expressed in the cochlear lateral wall, particularly in the suprastrial region.³² Light microscopy analysis shows that COX-1 is uniformly distributed in almost all cell types of the Corti organ, except the keratinized plates of columnar cells.³³ It is primarily associated with

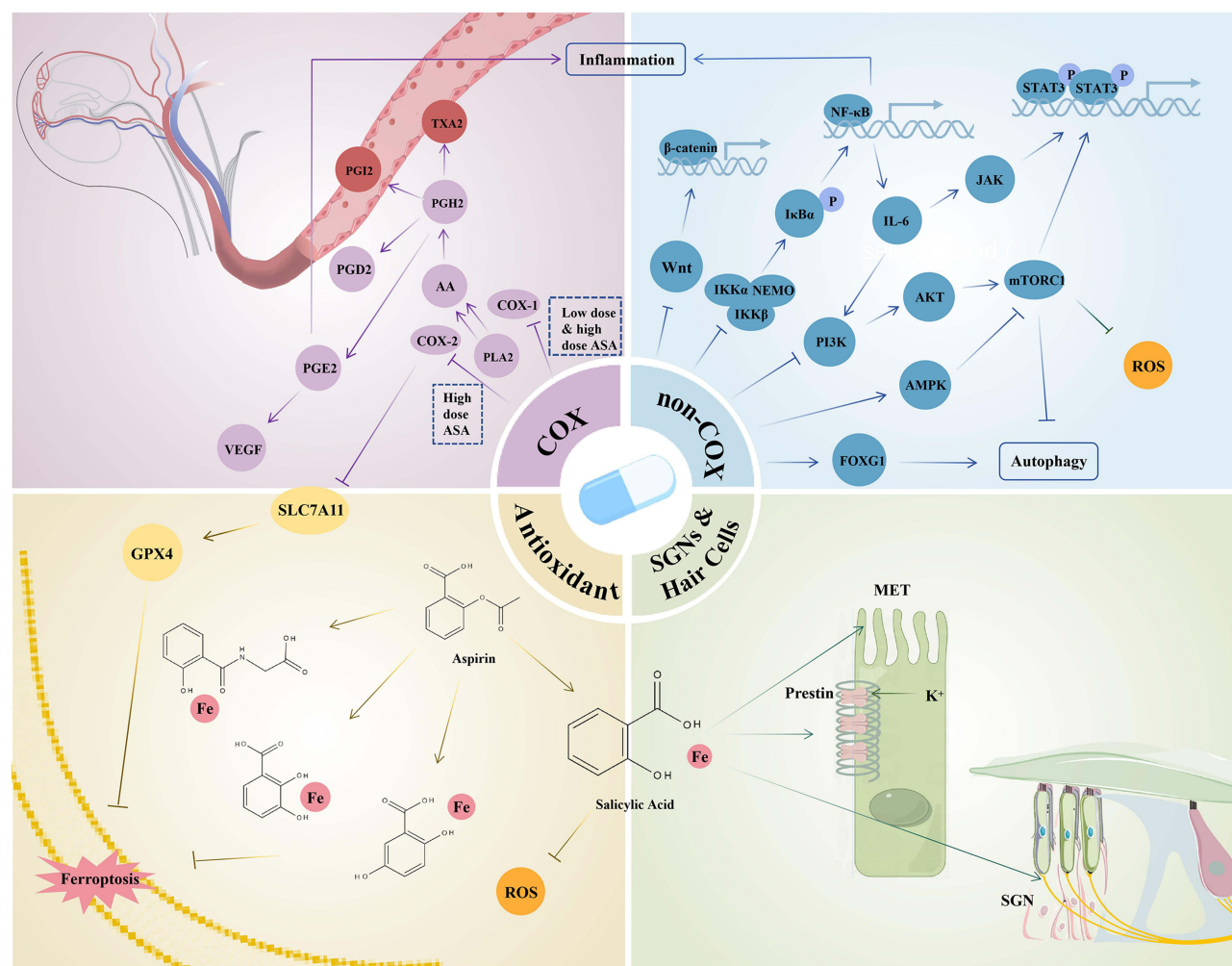


Figure 1 Aspirin affects hearing through four primary pathways: COX-dependent mechanism (purple box), COX-independent mechanism (blue box), antioxidation-mediated mechanism (yellow box), and prestin-mediated (green box) mechanism. Arrows (→) indicate activation, promotion, or downstream signaling; blunt ends (—) indicate inhibition or suppression.

maintaining gastrointestinal mucosal integrity, regulating platelet function, and modulating peripheral vascular resistance and blood flow distribution.³⁴ COX-1 is also more potently inhibited by aspirin than COX-2. In contrast, COX-2 is an inducible isozyme that is strongly expressed in response to pro-inflammatory stimuli and is found in the vascular endothelium, brain, kidney, and other organs.³⁵ In the Corti organ, COX-2 is expressed in all cell types, with particularly strong expression in the keratinocytes of Hensen cells, adjacent Deiters cells, and outer hair cells (OHCs).³³ Both COX-1 and COX-2 are detectable in the spiral ganglia.³³

Sound stimulation activates inner hair cells, triggering an increase in intracellular calcium ions (Ca^{2+}) concentration in fibrocytes via gap junctions. The elevated Ca^{2+} signal propagates through end-foot structures of the fibrocytes to adjacent vascular pericytes and endothelial cells. The rise in Ca^{2+} activates COX-1, which converts AA into vasoactive prostaglandins, such as prostaglandin E2 (PGE2). These prostaglandins diffuse into the perivascular space, inducing pericyte relaxation and consequent capillary dilation. The resulting vasodilation enhances local blood flow, thereby improving the delivery of oxygen and nutrients and facilitating the clearance of metabolic waste. This mechanism ensures the maintenance of normal auditory transduction during sound stimulation.³² Excessive noise causes constriction of cochlear blood vessels, reduces blood flow, and leads to ischemia and hypoxia in the inner ear.³⁶ Research indicates that exposure to moderate-intensity (70 dB - 90 dB) noise for 1 hour leads to a significant downregulation of COX-1 expression, particularly in various cell types, including Deiters' cells, inner and OHCs, and pillar cells. This reduction in

COX-1 may result in decreased synthesis of prostaglandins, thereby influencing glutamate release and reducing the sensitivity of the sensory system.³⁷ The alteration in COX-1 expression levels in response to noise exposure may represent a protective mechanism of the cochlea against noise overstimulation. This viewpoint has been confirmed by Hoshino et al, who demonstrated that the COX-1 inhibitor indomethacin reduces hearing loss in mice caused by exposure to 4 kHz pure tone noise at a sound pressure level of 128 dB for 4 hours.³⁸

COX-2 catalyzes the conversion of AA to PGE2,³⁹ with PGE2 and vascular endothelial growth factor (VEGF) constituting its principal downstream effectors.⁴⁰ Evidence indicated that both COX-2 and PGE2 are key mediators in the development of inflammatory disorders, including airway and pulmonary inflammation.^{41,42} The protein expression of COX-2 up-regulates within 7 days following noise exposure, with a particularly pronounced increase observed after 24 hours following exposure. Pharmacological inhibition of COX-2 by NS-398 mitigates noise-induced hearing loss and cochlear hair cell damage.⁴³ Cisplatin induces COX-2 expression, resulting in concomitant elevations of PGE2 and VEGF, which contribute to cochlear inflammation and vascular dysfunction, thereby exacerbating hearing loss.⁴⁴ Moreover, AAV1-mediated VEGFA165 gene therapy enhances cochlear vascular function by promoting pericyte survival and proliferation, attenuating noise-induced auditory deficits.⁴⁵ Noise exposure itself upregulates VEGF expression within cochlear vascular networks. In addition, aspirin regulates the involvement of SLC7A11/GPX4 in ferroptosis by inhibiting COX2.^{46,47} Ferroptosis has been found to lead to noise,⁴⁸ ototoxicity,^{49–51} and ARHL.⁵²

The dose-dependent inhibition of COX isoforms by aspirin underlies its dual effects on cochlear function. At low doses, aspirin preferentially acetylates platelet COX-1, irreversibly suppressing the production of TXA2, a potent vasoconstrictor and platelet aggregator,^{53,54} while largely sparing endothelial COX-2–derived prostacyclin (PGI2), a vasodilator and endogenous inhibitor of platelet activation in preclinical models.⁵⁵ This selective action preserves cochlear perfusion, and in the context of inner ear injury, the anti-inflammatory and cytoprotective effects of COX-2 inhibition outweigh any impact on vasomotor function.⁴³ Mechanistically consistent as these observations are, confirmatory clinical data supporting meaningful cochlear protection in humans are still lacking.

However, this delicate balance is disrupted by high aspirin doses. Systemic administration of high-dose aspirin (eg., ≥ 200 mg/kg in animal models) results in non-selective inhibition of both COX-1 and COX-2.⁵⁶ Inhibition of COX-2 in endothelial cells markedly reduces PGI2 synthesis, impairing its vasodilatory and antiplatelet effects. Without PGI2, TXA2–mediated platelet aggregation and vasoconstriction dominate, creating a prothrombotic environment.⁵⁷ In the sensitive cochlear vasculature, this shift accelerates intravascular thrombosis and substantially reduces blood flow in animal studies.⁵⁸ The resulting ischemia and hypoxia disrupt ion homeostasis and energy metabolism in hair cells and SGNs, which may lead to temporary hearing loss and tinnitus.⁵⁸ Although this pathway provides a biologically plausible explanation for aspirin related ototoxicity, direct translation to human ototoxicity has not been definitively confirmed in controlled clinical trials.

This pathophysiological cascade explains why high-dose aspirin exhibits ototoxicity by tipping the COX-dependent balance toward thrombosis and vasoconstriction. Aspirin therefore shows a dose-dependent dichotomy in cochlear vascular regulation: low doses reduce thrombosis and improve microcirculatory perfusion, whereas high doses promote thrombosis and ischemia. Given the suppression of PGI2 by high-dose aspirin, co-administration of AA with aspirin may offer a potential strategy to prevent or mitigate hearing loss.⁵⁹ However, this strategy remains preclinical and has not been validated in human clinical trials.

COX-Independent Mechanism

The COX-independent pathway encompasses multiple distinct sub-mechanisms, including NF-κB/STAT3 modulation, autophagy regulation, mechanoelectrical transducer (MET) channel inhibition, Prestin binding, SGN damage and antioxidant effects, which are detailed in the following subsections.

Aspirin exerts anti-inflammatory effects through a COX-independent mechanism, primarily by inhibiting the nuclear factor kappa B (NF-κB) and signal transducer and activator of transcription 3 (STAT3) pathways.^{60,61} When cells are stimulated by external factors such as inflammatory factors, the NEMO (IKK γ)/IKK α/IKK β kinase complex is activated. Subsequent phosphorylation and ubiquitin-mediated degradation of IκBs proteins liberate NF-κB dimers, predominantly the p50/p65 heterocomplex, which translocate into the nucleus to regulate gene transcription and modulate

inflammatory responses.⁶² NF- κ B is activated by noise, aging, and ototoxic drugs, driving inflammatory responses in cochlear hair cells.^{63–65} Aspirin has been reported to interact with IKK β , blocking the activation of the hub signaling complex of the NF- κ B signaling pathway, thereby inhibiting the expression of downstream inflammatory genes, such as IL-6.⁶⁶ Cellular exposure to stimuli (eg., noise, drugs) induces the secretion of cytokines such as interleukin-6 (IL-6).⁶⁷ The engagement of IL-6 with its cognate receptor activates receptor-associated Janus kinases (JAK), resulting in the phosphorylation of the transcription factor STAT3. Phosphorylation induces STAT3 dimerization and its subsequent nuclear import. In the nucleus, the STAT3 dimer binds to specific promoter regions, mediating the transcriptional activation of genes that promote cellular growth and survival. STAT3 regulates cell survival, differentiation, and microtubule dynamics in a cell- and function-specific manner within the cochlear sensory epithelium.^{68,69} RiboTag and single cell RNA sequencing shows that noise stimulation significantly induced upregulation of transcription factor STAT3 and its downstream genes, which are involved in immune and inflammatory responses.⁷⁰ Aspirin may reduce the generation and release of IL-6 in the cochlea from the source by inhibiting NF- κ B, thereby alleviating immune response and inhibiting ferroptosis through JAK/STAT3. However, this hypothesis has not yet been confirmed in the cochlea. This cascade is supported by mechanistic and preclinical studies but has not been validated in the human cochlea.

Aspirin has also been shown to induce forkhead box G1 (FOXG1) expression and autophagy, which reduces the production of reactive oxygen species (ROS) and apoptosis, thereby promoting the survival of aging hair cells.⁷¹ Mechanistically, autophagy may be triggered via adenosine monophosphate-activated protein kinase (AMPK) activation or phosphoinositide 3-kinase/protein kinase B (PI3K/AKT) inhibition, accompanied by suppression of mammalian target of rapamycin complex 1 (mTORC1).^{72–74} Nevertheless, AMPK signaling in the inner ear is highly complex and context-dependent. AMPK can exert either protective or injurious effects on cochlear hair cells depending on the type and duration of stress, whereas overactivation exacerbates injury.^{75,76} While these pathways are strongly supported in vitro and in animal models, the precise regulation of AMPK by aspirin in the human cochlea remains poorly defined, warranting further investigation.

MET, Prestin and SGNs

OHCs depend on two complementary molecular systems to enable normal hearing: MET channels located at the apical stereocilia and the motor protein Prestin (SLC26A5) distributed along the lateral membrane.⁷⁷ MET channels convert sound-induced mechanical forces into electrical receptor potentials, whereas prestin mediates voltage-dependent OHC electromotility that drives cochlear amplification.⁷⁸ Aspirin and its primary active metabolite salicylate modulate both targets through distinct and independent mechanisms, as established in preclinical and human pharmacodynamic studies.^{79,80}

MET channels serve as the primary entry route for ototoxic drugs such as aminoglycosides into hair cells.⁷⁹ Salicylate directly and reversibly blocks MET channel currents in a concentration-dependent manner, thereby reducing drug uptake and protecting hair cells from permanent damage.⁷⁹ This represents a major otoprotective mechanism that has often been underemphasized relative to effects on prestin.

In parallel, salicylate, the active metabolite of aspirin, binds directly to prestin by competing with chloride ions (Cl^-) at the protein's anion-binding pocket.⁸¹ This binding locks prestin in an intermediate conformational state, preventing the normal structural transitions required for OHC electromotility.^{82,83} As a result, salicylate inhibits the voltage-dependent mechanical activity of prestin, leading to reversible hearing impairment at high doses.⁸⁴

Studies in pigs have shown that long-term salicylic acid treatment (200–350 mg/kg/day for 2–3 weeks) leads to a compensatory upregulation of Prestin. After 14 consecutive days of administration, experimental data revealed increased distortion-product otoacoustic emission (DPOAE) amplitudes, along with elevated Prestin protein expression, further enhancing the OHCs' electrodynamics and improving cochlear function.⁸⁵ Despite the enhancement of DPOAE, prolonged exposure to high doses of salicylic acid can result in a permanent reduction in neural response amplitude, suggesting potential damage to SGNs.⁸⁶

One proposed mechanism is that salicylic acid potentiates N-methyl-D-aspartate (NMDA) receptor currents in SGNs, leading to glutamate excitotoxicity and irreversible neuronal injury.⁸⁶ Early-stage damage to cochlear synapses and auditory nerve fibers in the peripheral auditory system may also contribute.⁸⁷ This apparent paradox, in which the

mechanical function of the OHC is enhanced while neural output is diminished, is reflected in the modest, non-significant threshold shifts observed in salicylate-treated rats, typically ranging from only 5 to 10 dB.⁸⁵ Consequently, this type of neural damage may go undetected in standard hearing assessments, likely explaining its absence in earlier reports. The dichotomy between OHC functional enhancement and neuronal damage underlies the dual role of aspirin in auditory dysfunction.

Notably, aspirin exerts otoprotective effects primarily through blocking MET channels, whereas its reversible ototoxicity mainly arises from inhibiting prestin. These two independent and divergent actions highlight the distinct contributions of MET and prestin to the auditory effects of aspirin. Meanwhile, prolonged high-dose exposure can also cause permanent damage to SGNs via NMDA-mediated excitotoxicity, independent of both MET and prestin function. Together, these three parallel effects constitute the complex dual role of aspirin in the auditory system: protective effects via inhibition of MET channels at appropriate concentrations, and acute ototoxicity via prestin inhibition as well as potential neurotoxicity via SGN damage at high concentrations. All observations described in this section are limited to preclinical animal models, and no robust clinical or human *in vivo* evidence currently supports these findings; however, these mechanistic insights provide a theoretical basis and clinical guidance for future translational and clinical studies.

Antioxidant

Aspirin is an efficient hydroxyl radical scavenger at the biochemical level, with a reaction rate constant of $k = 3.6 \times 10^{10} \text{ M}^{-1} \text{ sec}^{-1}$, which is faster than several well-established antioxidants, such as ascorbate, glutathione, and cysteine.⁸⁸ This finding is derived from cell-free chemical assays. Most metabolites of aspirin have antioxidant and iron binding properties. At the cellular level, the active ingredient of aspirin is acetylsalicylic acid, which is an antioxidant that can clear ROS and protect hair cells from oxidative stress damage as demonstrated *in vitro*. In animal models, previous studies have observed that administration of salicylates reduces gentamicin-induced hair cell damage and ROS production in both reptiles and mammals.^{14–17} In addition, in noise-induced hearing loss, the combination therapy of salicylates and trolox significantly reduced auditory brainstem response defects in guinea pigs, decreased hair cell damage, and reduced the formation of ROS and reactive nitrogen species.⁸⁹

In addition to direct ROS scavenging, aspirin and its metabolites exert indirect antioxidant effects by modulating iron homeostasis. The following sections discuss iron chelation and metal ion metabolism as indirect mechanisms that mitigate oxidative stress through the inhibition of ROS-generating Fenton reactions. Aspirin metabolites have a high metal-chelating potential.²⁴ In particular, the four active compounds found in aspirin have all been widely reported to affect iron metabolism. Aspirin has been reported to possess weak iron-binding properties, and the formation of aspirin-iron complexes has been shown to exhibit pro-oxidant activity and toxicity in liver mitochondria.^{5,24} Other studies have similarly demonstrated that such iron complexes can promote oxidative stress and mitochondrial damage in the liver.^{90–92} In addition, aspirin modulates the expression of iron transport and storage proteins, as well as related metabolic pathways and physiological functions.²⁴ These effects on iron metabolism and chelation may represent a potential mechanism for hearing protection. For example, tert-butyl hydroperoxide (t-BHP) has been shown to induce iron accumulation and ferroptosis in HEI-OC1 cells and cochlear explants under oxidative stress conditions.⁹³ Moreover, 2,5-dihydroxybenzoic acid, a metabolite of the aminocarboxymuconate semialdehyde (ACMS) pathway, has been reported to protect against kanamycin-induced hair cell damage.⁹⁴

While these mechanistic studies provide a strong biological rationale, it is important to note that most derive from *in vitro* or animal models. The translational relevance of these pathways to human hearing loss, particularly at therapeutic aspirin doses, remains to be validated.

Aspirin and Hearing Loss

The relationship between aspirin and hearing is complex, involving both protective effects and potential ototoxicity. The impact of aspirin on hearing varies depending on underlying factors that contribute to hearing loss.

Aspirin and Age-Related Hearing Loss

Age-related hearing loss is common in the aging population, affecting communication and contributing to a worse quality of life.^{95,96}

Preclinical studies in animal models have shown that aspirin can inhibit systemic inflammatory biomarkers associated with aging.⁹⁷ In various mouse models, long-term aspirin use has been reported to extend lifespan and exert anti-aging effects.^{98,99} Mechanistically, aspirin treatment activates FOXG1 expression and the autophagy pathway, reducing ROS levels, inhibiting apoptosis, and ultimately promoting the survival of simulated aging hair cells and HEI-OC-1 cells.⁷¹ However, a multicenter, randomized, double-blind, placebo-controlled trial in healthy community-dwelling adults aged 70 years or older in Australia found that daily administration of 100 mg of enteric-coated aspirin did not slow or accelerate the progression of hearing loss^{20,97,100} (Table 1).

Although aspirin has shown promise in basic research by reducing systemic inflammation, activating autophagy, lowering oxidative stress, and extending lifespan in animal models, these effects have not translated to humans with ARHL. The negative results of the Aspirin in Reducing Events in the Elderly (ASPREE) hearing study suggest that ARHL involves more complex mechanisms than inflammation or oxidative stress alone. Such factors may include cochlear microcirculation disorders, synaptic degeneration, and central auditory pathway changes. The anti-inflammatory and antiplatelet effects of low-dose aspirin may be insufficient to counter this multifactorial degeneration. Differences in dosage and formulation further complicate translation from animals to humans. However, the external validity of the ASPREE hearing study is limited by a relatively homogeneous cohort and an insufficient sample size. Consequently, the results may not be generalizable to broader populations and are inadequate to draw conclusive evidence on the association between aspirin use and the progression of age-related hearing loss. In addition, aspirin has the potential to regulate skeletal muscle aging, improve intestinal homeostasis, and delay aging. Therefore, further research is required to confirm its potential in preventing ARHL.^{101,102}

Aspirin and Ototoxic Hearing Loss

With the widespread use of ototoxic agents, including chemotherapy drugs and aminoglycoside antibiotics, drug-induced hearing loss has emerged as a significant worldwide health problem.¹⁰³ As many as 93% of individuals treated with cisplatin chemotherapy may experience hearing impairment, which seriously affects the quality of life among cancer survivors.¹⁰⁴

Salicylic acid has been reported to protect against gentamicin-induced hepatotoxicity and nephrotoxicity both in rabbits and rats.^{105–108} Numerous studies have also found that aspirin can reduce aminoglycoside-induced hearing loss.^{16,17,19,109} Sha et al and our team reported that moderate concentrations of aspirin (975 ± 212 mg) can significantly reduce the risk of gentamicin-induced hearing loss.^{17,19} A prospective, randomized, double-blind trial in 60 patients treated with gentamicin found that patients treated with 1.5 g/day of aspirin (500 mg every 8 hours) experienced significant hearing improvement compared to the placebo group.¹⁶ Aspirin may protect against aminoglycoside-induced ototoxicity through several potential mechanisms. It can competitively inhibit MET channels, thereby reducing gentamicin uptake into hair cells, and may also interfere with prestin function, indirectly mitigating the toxic effects on cochlear amplification.⁷⁹ Additionally, aspirin's antioxidant properties may alleviate oxidative stress-induced damage.⁷⁹

Table 1 Studies Evaluating the Impact of Low Aspirin Doses on the Progression of Age-Related Hearing Loss within 3 Years

Study Design (Sample Size)	Patient Description	ASA Regimen Evaluated	Hearing Evaluation Results	Conclusion	Ref.
Randomized controlled study (n=1,260)	Healthy Australians, Age > 70 years	ASA 100mg/d vs. placebo	Changes in mean hearing thresholds between baseline and 3 years after treatment, ASA vs placebo: 4FA 0.31 dB (95% CI, -0.72 to 1.34 dB); SRT: -0.1 [95% CI, -1.2 to 1.0]; P = 0.86)	Low-dose aspirin has no significant effect on the progression of ARHL within 3 years	[97]

Abbreviations: ASA, aspirin; 4FA, the four-frequency average (defined as the mean air-conduction hearing threshold at 0.5, 1, 2, and 4 kHz in the better-hearing ear); SRT, speech reception threshold; CI, confidence interval; ARHL:age-related hearing loss.

In preclinical studies, systemic administration of aspirin in rats and mice conferred protection against cisplatin-induced HC damage and hearing loss.^{110,111} However, clinical trials have consistently failed to demonstrate a protective effect of aspirin against cisplatin-related ototoxicity^{21,112} (Table 2). These findings suggest that cisplatin- and gentamicin-induced ototoxicity involve distinct pathomechanisms and may require distinct therapeutic strategies. Several mechanistic differences may explain why aspirin protects against gentamicin- but not cisplatin-induced hearing loss. First, gentamicin enters hair cells primarily through MET channels, and aspirin has been shown to competitively inhibit these channels, thereby reducing gentamicin uptake into hair cells.⁷⁹ In contrast, cisplatin enters hair cells mainly via copper transporter Ctr1 and organic cation transporter OCT2,¹¹³ pathways that are not affected by aspirin. Second, while both drugs induce oxidative stress, cisplatin causes additional DNA crosslinking and activation of distinct apoptotic pathways (eg., p53-mediated apoptosis),¹¹⁴ which are not effectively counteracted by aspirin's antioxidant properties alone. Third, differences between animal models and clinical trials should also be considered. Preclinical studies often use younger, healthier animals with no comorbidities, and may employ higher aspirin doses or different administration schedules than those feasible in human patients.¹¹⁵ Additionally, animal studies typically assess hearing over short-term follow-up periods, whereas clinical trials involve longer observation times and more heterogeneous patient populations, which may mask or attenuate protective effects.

Aspirin and Noise-Induced Hearing Loss

Moderate noise exposure in occupational and recreational environments can also cause hearing loss. It is estimated that patients with noise-induced hearing loss account for about 10% of patients in otolaryngology clinics.¹¹⁷

Animal studies have shown that the combination of aspirin and noise at high doses (Male rat: 200 mg/kg twice daily for 8 days, followed by 125 mg/kg for 10 days, Female rats: 200 mg/kg twice daily for 8 days, followed by 150 mg/kg for 10 days) causes greater permanent hair cell loss than noise alone.¹¹⁸ However, a study on six chicks showed that aspirin did not have a synergistic effect with noise.¹¹⁹ Research shows that administering 300 or 350 mg/kg aspirin to mice reduced the hearing sound threshold caused by noise.¹²⁰ In clinical practice, three small-scale studies evaluated

Table 2 Studies Evaluating the Effect of Aspirin on Drug-Induced Hearing Loss

Study Design (Sample Size)	Patient Description	ASA Regimen Evaluated	Hearing Evaluation Results	Conclusion	Ref.
Randomized double-blind placebo-controlled study (n =195)	ASA: 76 males and 13 females Placebo: 95 males and 11 females treated with gentamicin 80–160 mg intravenously twice daily	ASA 1 g Q8h ×14d	ASA: 3% (3/89) had HL 15 dB at 6000 Hz and at 8000 Hz P =0.013 Placebo:13% (14/106) had HL 15 dB at 6000 Hz and at 8000 Hz	ASA is effective in reducing ototoxicity from treatment with gentamicin	[17,19]
Randomized double-blind controlled study (n =60)	Age > 18 years, Scheduled gentamicin 80 mg 3 times daily intravenously Exclusion criteria: preexisting HL, systemic disease, and pregnancy	Experimental group: ASA 1.5 g/d × 7 d Control group: given placebo	ASA: 3% (1/30) had HL 15 dB at 4000 Hz (P =0.001), 3% (1/30) had HL 15 dB at 8000 Hz (P =0.0014) Placebo: 36% (11/30) had HL 15 dB at 4000 Hz (P =0.001), 20% (6/30) had HL 15 dB at 8000 Hz P=0.0014	ASA was associated with better PTAs	[16]
Phase II double-blind, randomised controlled trial (n=94)	Adults (≥18 years) receiving cisplatin-based chemo, planned cumulative cisplatin dose ≥200 mg/m ²	ASA 975 mg tid (+ omeprazole) vs. placebo per chemo cycle	Mean HL (6+8 kHz): ASA arm 49.0 dB (SD 61.41) vs Placebo 36.0 dB (SD 50.85); LS mean difference 9.38 dB (60% CI: – 1.45 to 20.22; p=0.233)	Aspirin did not protect from cisplatin-related ototoxicity.	[21,112]

Notes: Partial data in this table are adapted from Kyle et al, 2015.¹¹⁶

Abbreviations: ASA, aspirin; HL, hearing loss; PTA, pure-tone average; CI, confidence interval; LS, least squares.

whether simultaneous intake of aspirin would enhance the impact of noise on hearing after noise exposure, with mixed results. Two of these studies showed that the administration of aspirin at doses of 1.95 to 3.9 g per day for 0.5 to 4 days has been shown to potentiate noise-induced temporary threshold shifts, with increases of 18–27 dB compared to 14 dB without aspirin.^{121,122} These findings suggest that the administration of aspirin should be avoided before anticipated acoustic overstimulation. In contrast, a trial using a single 1-g dose administered one hour before noise exposure over five sessions did not demonstrate a synergistic interaction between aspirin and noise¹²³ (Table 3).

Previous clinical studies were limited by small sample sizes ($n = 4$ –11), resulting in insufficient statistical power to reliably assess the interaction between aspirin and noise and increasing the risk of type I or II errors. Small cohorts also limit control over potential confounders, including baseline hearing, noise exposure, and overall health, which may contribute to conflicting results. Future research should address these limitations through large, well-controlled randomized clinical trials to clarify the dose-dependent effects of aspirin on noise-induced hearing loss.

Aspirin and Tinnitus

Evidence of high-dose salicylate-induced tinnitus and hearing impairment was derived from clinical studies.¹²⁴ Studies have found a linear correlation between the loudness of tinnitus and the blood concentration of salicylates. Tinnitus typically occurs when plasma salicylate concentrations fall below 100 mg/L, whereas levels near 300 mg/L produce a perceived loudness of approximately 60 dB.⁵⁸ Aspirin-induced tinnitus has been attributed to several mechanisms, including cochlear hearing loss leading to enhanced central auditory gain, alterations in neural plasticity within the central auditory pathways, and excessive secretion of corticosterone.^{125–127} On the other hand, a clinical trial showed that daily administration of 1.2 g of aspirin could suppress tinnitus caused by spontaneous otoacoustic emissions (SOAE). However, as SOAE accounts for only about 4% of tinnitus cases, aspirin cannot be considered a broadly applicable palliative treatment.¹²⁸

Table 3 Studies Evaluating the Effect of Aspirin on Noise-Induced Hearing Loss

Study Design (Sample Size)	Patient Description	ASA Regimen Evaluated	Hearing Evaluation Results	Reversible	Conclusion	Ref.
Prospective crossover trial ($n = 11$)	11 males, 19–23 years old, with normal hearing	ASA 3.9 g/d $\times$ 4d All sessions with intensity necessary for 10 min 2500 Hz tone to produce ~12 dB of TTS	ASA with exposure to intense noise produced HL ~10–15 dB greater than that produced by exposure to the intense sound alone.	NR	Moderate doses of ASA may increase the risk of noise-induced hearing loss.	[121]
Prospective cohort trial ($n = 4$)	4 volunteers (age, gender unspecified) exposed to 10 min 2500 Hz tone at intensity necessary to produce 14 dB TTS at baseline	ASA 1.95 or 3.9 g/d $\times$ 0.5–2.75 d	Exposures that ordinarily produce 14 dB TTS produce 18–27dB instead: the effect is dose and duration-dependent.	All hearing losses were reversed at 24–48 h after the exposure	ASA leads to a worse temporary threshold shift after noise	[122]
Prospective cohort trial ($n = 8$)	Young males (mean age 19.3 years) with normal hearing (250 Hz–8 kHz) exposed to noise (one-third octave band filtered noise with 2 kHz CF at 105 dB SPL for 10 min)	1 g ASA was provided 1 h before noise exposure in 5 sessions Control: no ASA	ASA: TTS was reduced in the fourth session only. ASA did not affect the TTS.	—	ASA has no synergistic effect with noise	[123]

Notes: Data in this table are adapted from Kyle et al, 2015.¹¹⁶

Abbreviations: ASA, aspirin; TTS, temporary threshold shift; NR, not reported.

Relationship Between Aspirin Dose and Hearing Loss

This review summarizes reports from various clinical studies (Table 4). The clinical studies evaluated the association of aspirin dose with hearing loss.

The impact of aspirin dosage on hearing loss remains controversial. Studies on rats and mice have shown that high levels of the active ingredient sodium salicylate (SS) (>200 mg/kg) can lead to hearing impairment in rats and mice,^{136,137} and result in long-term inner ear damage.¹³⁸ Nevertheless, lower doses of SS (200 mg/kg/day) can cause tinnitus but not hearing impairment.⁸⁷

A systematic review of aspirin and sensorineural hearing loss (SNHL) in 37 clinical studies found that high-dose aspirin (especially ≥ 1.95 g/d) resulted in a reversible decrease in hearing threshold (4–112 dB).¹¹⁶ However, due to the small sample size and insufficient control of confounding factors, it is impossible to answer the core clinical question of whether long-term low-dose medication is safe. The hearing loss was dose-dependent on aspirin. In patients with rheumatoid arthritis or connective tissue disease, high doses of aspirin (3.6–9.0 g/d) resulted in temporary hearing loss. Conway et al compared patients with SSNHL who had used aspirin before symptom onset ($n = 38$) with those who had not ($n = 110$). Aspirin use, irrespective of dosage, was associated with poorer auditory outcomes than in patients who did not develop SNHL after using aspirin.²⁹ Conversely, a retrospective study by Rife et al showed that a high dose of aspirin (80–100 mg/kg/day, three times a day, for 7–12 days) did not cause SNHL in children ($n = 31$) with Kawasaki disease.¹³² Liu et al demonstrated that in patients with type 2 diabetes, administration of low-doses of aspirin (cumulative defined daily dose (cDDD) <1072) did not protect against hearing loss, whereas a higher cumulative dose (cDDD ≥ 1072) had a protective effect against SSNHL.¹²⁹

When interpreting the available clinical and animal data collectively, a dose-dependent pattern emerges, though with considerable caveats. Short-term use of low-dose aspirin (81–325 mg/day, mainly for antiplatelet therapy) with a cDDD of less than 700 has not been consistently associated with hearing protection in available studies. It is important to note that this observation is derived from heterogeneous sources and should be considered hypothesis-generating rather than definitive clinical guidance. At these doses, the anti-inflammatory and antithrombotic effects of aspirin may be insufficient to provide sustained microcirculatory protection. Long-term exposure to low-dose aspirin with a cDDD greater than 1072 has not demonstrated clear ototoxicity in general populations and, in one retrospective study of diabetic patients, was linked to a reduced risk of sudden hearing loss. However, this finding requires prospective confirmation.¹²⁹

Table 4 Impact of Different Aspirin Doses on Patient Hearing (2013–2025)

Study Design (Sample Size)	Patient Description	ASA Regimen Evaluated	Hearing Evaluation	Conclusion	Reversible	Ref.
Cohort study (n=103314)	Age > 18 years, T2D patients	cDDD of aspirin ≥ 1072 ; cDDD <1072	The unadjusted model between overall aspirin use and SSNHL risk, HR = 1.00, 95% CI = 0.86–1.15; $p = 0.9683$; Q1 (cDDD between 28 and 300):aHR = 1.26, 95% CI = 1.01–1.57, $p = 0.0405$; Q2 (cDDD between 301 and 700):aHR = 1.43, 95% CI = 1.16–1.75, $p = 0.0006$ Q4 (cDDD greater than 1072) (aHR =0.46, 95% CI = 0.34–0.62; $p < 0.0001$)	There is no statistically significant correlation between aspirin use and SSNHL. Elevated aspirin dosages have a protective effect. There is no difference between daily cDDD<1 and ≥ 1	NR	[129]

(Continued)

Table 4 (Continued).

Study Design (Sample Size)	Patient Description	ASA Regimen Evaluated	Hearing Evaluation	Conclusion	Reversible	Ref.
Retrospective study n=132	Deliberate aspirin overdoses in patients	>3000mg	35 (28%) patients presented with tinnitus	Excessive use of ASA did not result in deafness in patients suffering from tinnitus	-	[130]
Prospective cohort study (n =3136)	Age 25–42 years	At least weekly aspirin use (including baby aspirin)	The estimates in the Propensity Score Matching for the aspirin-treated group were 0.48 (95% CI: 0.06–0.90) for averaged ear analysis and 0.48 (95% CI: 0.05–0.91) for both ear analysis	Hearing deterioration at 500 Hz is greater among those who take aspirin weekly. All other hearing frequencies are not affected.	NR	[131]
Retrospective study (n=31)	Kawasaki disease children: mean age 43.01 months	Acute phase: within 10 days of the start of fever or 24–48 hours after defervescence, ASA 80–100 mg/kg/day, three times a day; Subacute phase: ASA 3–5 mg/kg/day, once daily for 8 weeks	None of the patients had SNHL before and 48 hours after starting treatment, and at least 20 days after the end of treatment (completed by 17 patients (54.84%))	High-dose ASA did not cause SNHL	-	[132]
Cross-sectional comparative study (n=403)	ASA: 182 heart disease patients who have been taking aspirin for a long time (≥1 year) Control: 221 non-ASA subjects	75mg/d (≥1 year)	36.19% (115/182) of ASA subjects presented with tinnitus. 56.11% (124/221) of controls presented with tinnitus. 65.93% (120/182) of ASA subjects presented with SNHL. 57.01% (126/221) controls presented with SNHL (p-value= 0.68).	When considering age, hypertension, diabetes, and other confounding factors, the antiplatelet dose of aspirin was not associated with any hearing problems, such as tinnitus and hearing loss.	NR	[133]
Case study n=1	25-year-old female	500mg×70 tablets=35000mg	Tinnitus 2.5 hours after administration, but no hearing loss	Acute high-dose ASA did not cause hearing loss, only tinnitus	-	[134]
Case study n=1	46-year-old female	325 mg × 200 tablets=65000 mg	Bilateral symmetric SNHL with a threshold of 50 dB HL	ASA acute intoxication is reversible	Reversible	[58]
Case study n=1	27-year-old male	100 mg × 100 tablets=10000 mg	80% right: PTA 45 dB (average of 0.5 kHz, 1 kHz, 2 kHz, and 3 kHz) speech discrimination score: 76%, left: PTA 41 dB, speech discrimination score: 80%	ASA acute intoxication is reversible	Reversible	[135]

Abbreviations: ASA, aspirin; T2D: type 2 diabetes, cDDD, cumulative defined daily dose; SSNHL, sudden sensorineural hearing loss; SNHL, sensorineural hearing loss; HL: hearing loss; NR, not reported; PTA, pure-tone average; CI, confidence interval; Q, quartiles; aHR, adjusted hazard ratio.

Moderate doses of aspirin (325 mg–1.95 g/day), commonly used to treat pain or fever, may cause tinnitus without causing hearing loss. Animal studies indicate that moderate doses can damage cochlear synapses and SGNs without affecting the number or morphology of OHCs.⁸⁷ However, clinical evidence in humans is limited, and individual responses can vary. Short-term use of high-dose aspirin (>1.95 g/day, used for anti-inflammatory or rheumatic therapy) can cause temporary hearing threshold shifts, high-frequency hearing loss, and tinnitus, which are usually resolved after discontinuation of treatment. The ototoxicity may be caused by PGI₂-mediated thrombosis or activation of the NMDA receptors in the SGNs.^{57,86} Although long-term high-dose use may cause nerve damage, it does not produce significant hearing loss, possibly due to compensatory neural changes. In some cases, the combination of high-dose aspirin with gentamicin may even have protective effects.^{16,17,19,109} Variability in aspirin-induced SNHL may depend on age, underlying disease, or comorbidities such as coronary artery disease, stroke, systemic lupus erythematosus, or antiphospholipid syndrome.¹³⁹

Overall, these findings indicate that the dose-effect relationship of aspirin is complex, and aspirin has a dual effect on different types of hearing loss (Table 5). The treatment by adjusting the dose or extending dosing intervals may maximize cochlear protection while minimizing neurotoxicity. Individualized strategies should be considered for high-risk patients, particularly those with pre-existing hearing loss. It is critical to emphasize that the dose-response relationships summarized above derive from heterogeneous sources, including animal studies, case reports, small cohort studies, and only a few randomized trials. Many of the apparent protective or toxic effects have not been replicated in large-scale, high-quality clinical studies. Therefore, the patterns described should be viewed as hypothesis-generating rather than as definitive clinical guidance.

Aspirin and Nanotechnology

Current Status of Nanodelivery Systems

In basic research, aspirin is mostly administered intravenously, whereas in clinical practice, it is generally given orally. The BLB restricts entry of most molecules into the inner ear. Following systemic administration, less than 1% of even small, lipophilic drugs can cross the BLB, severely limiting inner ear drug efficacy.¹⁴⁹ Therefore, the achievement of

Table 5 Dose-Response Relationship Diagram

Dose Range	Age-Related Hearing Loss	Ototoxic Hearing Loss	Noise-induced Hearing Loss	SSNHL in Subjects without Inflammatory Conditions	SSNHL in Subjects with Underlying Diseases
81–325 mg/d	Ineffective ^{20,97,100}	Not reported	NR	Minor damage ¹³¹	Protective (cDDD ≥ 1072 has a protective effect on the hearing of patients with diabetes) ¹²⁹
325 mg –1.95 g/d	NR	Protecting against gentamicin-induced hearing loss, ^{17,19} ineffective with cisplatin-induced hearing loss ^{21,112}	Ineffective ¹²³	May cause tinnitus ¹²⁸	Reversible hearing loss (dosing frequency NR) ¹⁴⁰
>1.95 g/d	NR	NR	Damaging ^{121,122}	Short-term temporary threshold shift is dose-dependent and reversible ^{116,141–143}	Damaging (Rheumatoid Arthritis and Connective Tissue Diseases); ^{144–148} Ineffective (Kawasaki disease) ¹³²

Note: The dose-response relationships summarized in this table are derived from heterogeneous sources, including case reports, and a limited number of randomized trials. The variability across studies may not be fully captured in this simplified summary. Readers are encouraged to consult the original publications for detailed study characteristics.

Abbreviation: NR, not reported.

therapeutic concentration often requires high systemic doses, increasing the risk of gastrointestinal, hepatic, and renal toxicity.¹⁵⁰ In addition, aspirin ototoxicity is largely due to its damage to SGNs, thus masking its advantages in enhancing the OHCs' electrodynamics and improving cochlear function. Local drug delivery has emerged as a promising alternative,¹⁵¹ enabling precise targeting of OHCs while reducing aspirin-induced SGN damage and enhancing hearing protection.

Aspirin can theoretically be delivered via the Eustachian tube for absorption through the round window membrane or directly into the inner ear via the round or oval window using cochlear implants, osmotic pumps, or perfusion systems.^{152–163} Intratympanic injection is the most commonly used clinical method,^{164,165} allowing higher inner ear drug concentrations by prolonging middle ear residence and avoiding first-pass metabolism.^{166–168} Combining aspirin with sustained-release systems such as hydrogels can effectively mitigate its ototoxicity, particularly for long-term protection against noise-induced hearing loss or age-related hearing loss.

Current nanomaterials employed for inner ear applications include hydrogels, cubosome nanoparticles, liposomes, polymer nanoparticles, silica nanoparticles, and superparamagnetic iron oxide nanoparticles (SPIONs)¹⁶⁹ (Figure 2). These technologies hold significant potential to improve drug delivery efficiency and treat specific pathologies.

Recent advances have highlighted the potential of nanomaterials for drug delivery in otology. One study employed a PTH/NPs composite emulsification method to prepare hydroxyl-modified polylactic acid (PLGA-SH) nanoparticles loaded with teriparatide (PTH1-34) (PLGA-SH NPs), and utilized the fusion peptide LR27, comprising the targeting

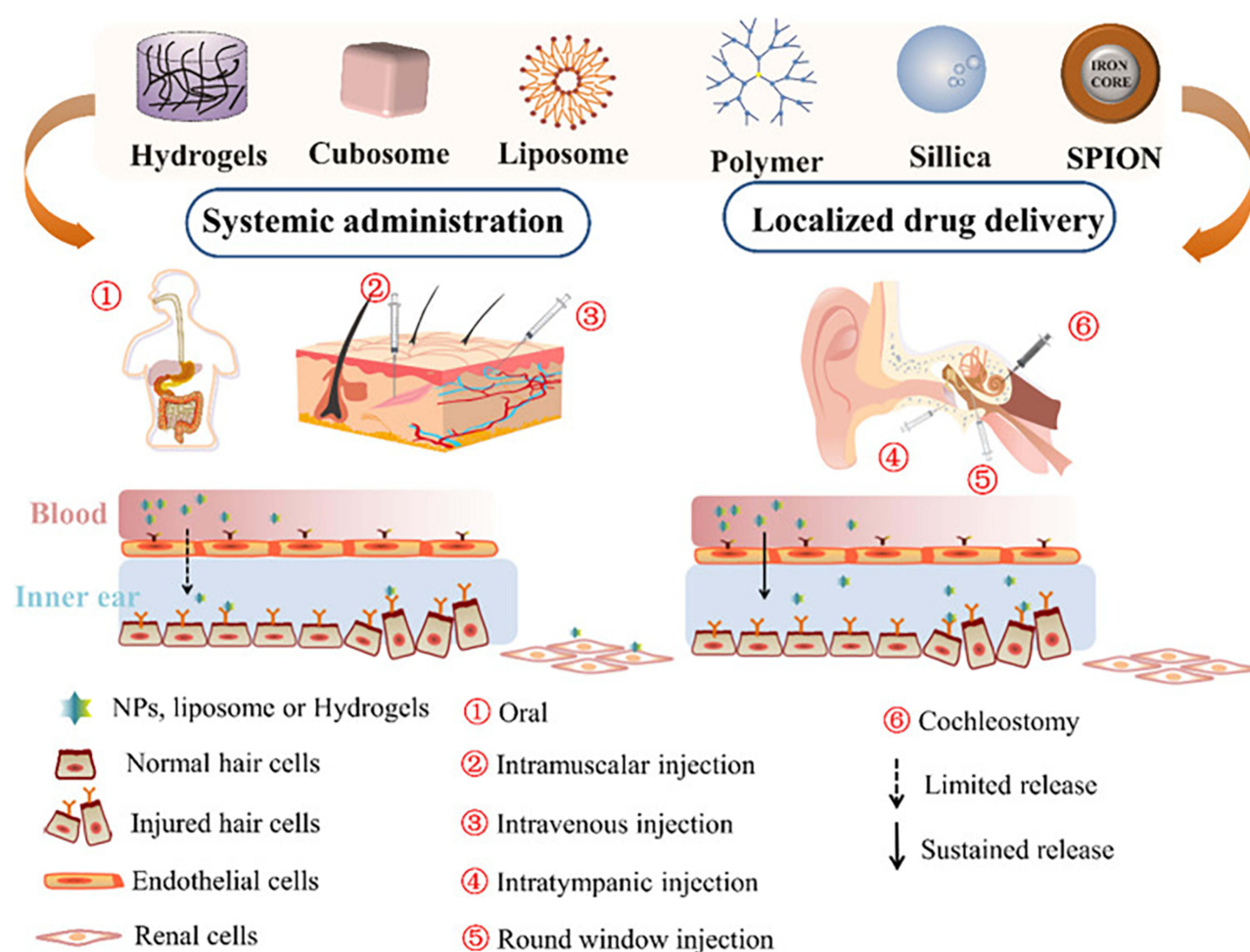


Figure 2 Nanomaterials and multiple administration methods.¹⁶⁹ Reprinted from Lin Q, Guo Q, Zhu M, et al, *Frontiers in Bioengineering and Biotechnology*, 2022, 9, 809443. Licensed under Creative Commons Attribution 4.0 International License (CC BY 4.0).

Abbreviations: SPIONs: superparamagnetic iron oxide nanoparticles; NPs: nanoparticle.

peptide A665 and cell-penetrating peptide Arg8, to target hair cells and enhance round window membrane permeability. A multifunctional thermosensitive nanodelivery system (Gel PTH/LRNPs) based on thermosensitive hydrogels enabled sustained release. In a noise-exposed mouse model, auditory bulb injection of this system effectively protected hearing and hair cells, with its mechanism linked to antioxidant and anti-apoptotic effects.¹⁷⁰ Similar studies further confirm that nanomaterial-based delivery systems improve drug-induced hearing loss and ARHL by enhancing BLB penetration, reducing systemic toxicity, and targeting damaged cochlear cells.^{171,172}

Aspirin Prospects and Optimization Strategies for Aspirin Nanodelivery Systems in Otology

In recent years, aspirin has been successfully incorporated into various nanodelivery systems, demonstrating broad therapeutic potential in antithrombotic, anti-tumor, anti-inflammatory, and ophthalmic applications. Huang et al prepared chitosan/sodium tripolyphosphate nanoparticles via ionic gelation (ASA-RGD-CS@TPP), which specifically target activated platelets, enabling efficient aspirin release, enhancing antithrombotic effects in rats, and reducing bleeding risk.¹⁷³ Albumin nanoparticles carrying aspirin exhibit good biocompatibility and sustained release in ophthalmic administration. Aspirin free drug releases 75% within 30 minutes, whereas nanoparticle formulations achieve 90% release over 72 hours, with a 50% release time of approximately 12 hours. Incorporation of xanthan gum further slowed release, extending 50% release to 40 hours and 100% release to 90 hours, offering a non-invasive strategy with high patient compliance for local treatment of diabetic retinopathy.¹⁷⁴

The successful integration of aspirin into nanodelivery systems for other therapeutic areas, including antithrombotic, anti-inflammatory, and anticancer applications, provides valuable insights for inner ear delivery.^{173–178} Previous studies have confirmed that the aspirin nanocarrier system is a simple, safe, and effective targeted treatment strategy (Figure 3).

The integration of aspirin with nanodelivery systems holds significant promise for applications in otology. Based on the pharmacokinetic limitations of aspirin and the molecular mechanisms underlying its dual effects on hearing, four optimization strategies warrant particular attention: ROS-responsive delivery, targeted delivery to cochlear structures, sustained-release formulations and combination therapy.

Noise exposure and ototoxic drugs induce elevated levels of ROS in the cochlea, which play a central role in hair cell damage and hearing loss.^{179–181} ROS-responsive nanodelivery systems¹⁸² represent a promising approach for treating noise-induced and drug-induced hearing loss. By combining cleavable linkers such as thioketal or diselenide and peroxalate ester bonds, aspirin can be released on-demand in inflamed cochlear tissue.^{183,184} Additionally, poly (propylene sulfide)₁₂₀ (PPS₁₂₀) has been shown to scavenge ROS and convert to poly (propylene sulfoxide)₁₂₀ in a ROS-rich environment, leading to carrier disintegration and rapid drug release with concurrent anti-inflammatory and antioxidant effects.¹⁸² Such strategies thereby minimize systemic exposure and reduce off-target toxicity while maximizing the antioxidant effect of aspirin at the site of injury.

To achieve selective accumulation of aspirin at the desired site of action, surface functionalization of nanoparticles with targeting ligands is essential. Two primary targets are relevant for hearing protection: (1) the stria vascularis, where aspirin may exert anti-inflammatory and antithrombotic effects via COX inhibition to improve cochlear microcirculation.^{41,42} The aspirin accumulation and retention in vascular regions could be increased by attaching stria vascularis ligand-specific markers, such as Claudin 11, KCNQ1, NKCC1 or KCNJ10,¹⁸⁵ to the surface of nanoparticles. (2) OHCs, where prestin serves as a unique molecular marker.¹⁸⁶ Our group has successfully designed a novel prestin-targeting peptide sequence, LS19, which can be utilized for targeted delivery of aspirin to OHCs.¹⁸⁶

To potentially mitigate ototoxicity associated with high-dose aspirin, sustained-release formulations represent a promising strategy that warrants systematic evaluation in preclinical models before clinical application. Encapsulation of aspirin in biodegradable polymers such as PLGA enables controlled drug release over days to weeks, with release kinetics tunable by adjusting the lactide-to-glycolide ratio.¹⁸⁷ Alternatively, incorporation of aspirin-loaded nanoparticles into thermosensitive hydrogels allows for a single intratympanic injection that forms a semi-solid depot at body temperature, extending the release duration.¹⁸⁸ This dual-layer sustained-release strategy effectively mitigates aspirin-induced ototoxicity while providing long-lasting protection against NIHL or ARHL.

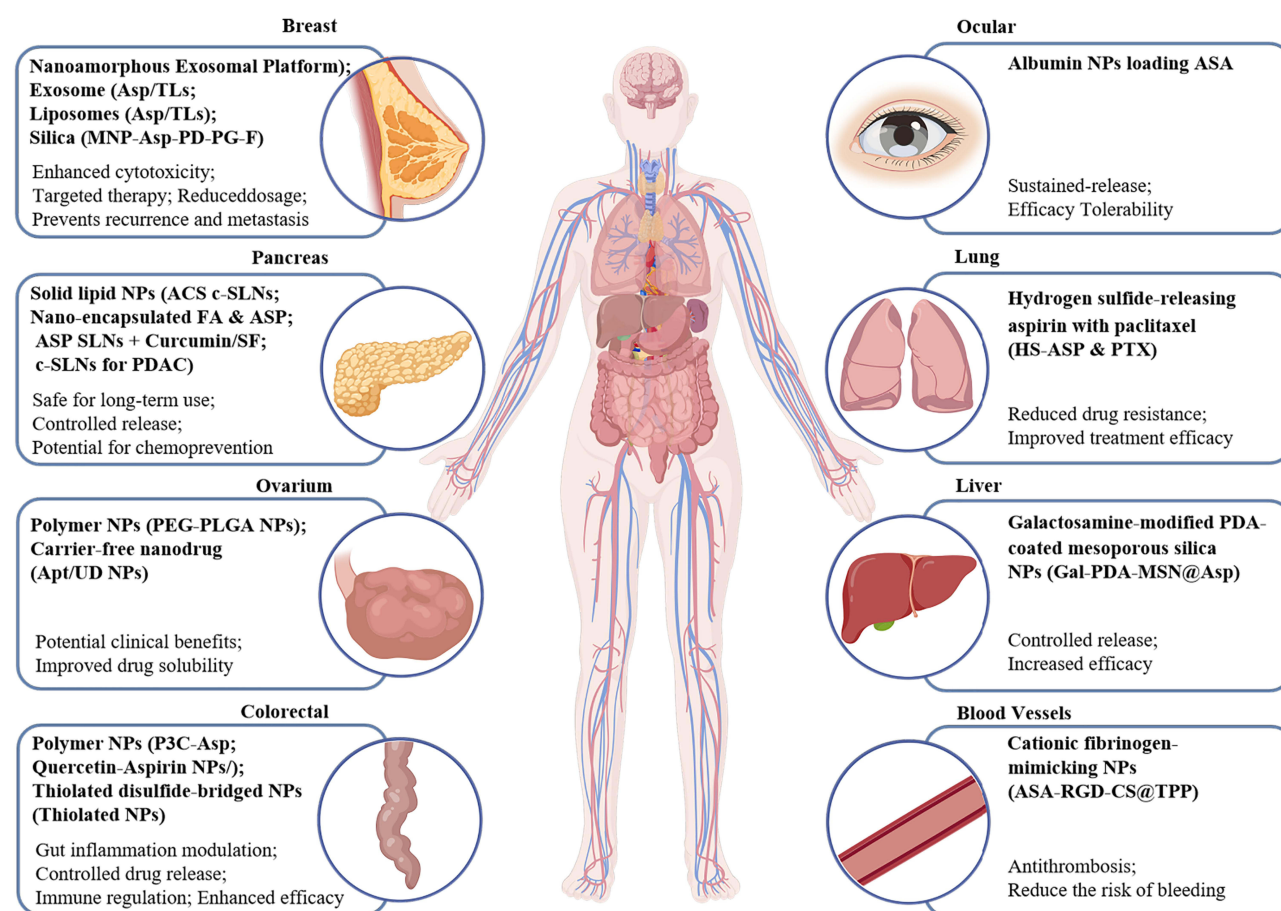


Figure 3 The use of aspirin in nano delivery systems in different tissues. This image depicts a nano delivery system constructed using different nanomaterials to carry aspirin in different tissues of the human body, as well as its effectiveness. The human body diagram highlights the widespread use and advantages of aspirin nanocarrier systems. **Abbreviations:** Asp/ASP/ASA, aspirin; TLS, Thermoresponsive liposomes; MNP, Mesoporous silica nanoparticles; PG, Polymer polyethylene glycol; PD, Polydopamine; F, Folic acid; ACS, aspirin curcumin and free sulforaphane; c-SLNs, Chitosan coated solid-lipid nanoparticle; FA, Folic acid-positive; SLNs, Solid-lipid nanoparticle; SF, Sulforaphane; PDAC, Pancreatic ductal adenocarcinoma; PEG, Polyethylene Glycol; PLGA, Poly (Lactic-co-Glycolic Acid); NPs, Nanoparticles; Apt, Aptamer; UD, Ultra-dispersed; P3C-Asp, A multifunctional oral dextran-aspirin nanomedicine; HS, Hydrogen sulfide; PTX, Paclitaxel; Gal, Galactosamine; PDA, polydopamine; MSN, Mesoporous silica nanoparticles; RGD, Arginine-glycine-aspartic acid; CS, Chitosan; TPP, Triphosphosphate.

Hearing loss pathogenesis involves multiple interconnected pathways, including oxidative stress, inflammation, and ferroptosis.^{50,189} Co-delivery of aspirin with other otoprotective agents using nanoparticle platforms may achieve synergistic effects. For example, combining aspirin with ferroptosis inhibitors (eg., Ferrostatin-1) could simultaneously inhibit COX-mediated inflammation and iron-dependent cell death.^{46,47} Similarly, co-administration with antioxidants (eg., berberine, curcumin) may provide enhanced protection against hair cell damage.^{182,190} Nanotechnology-based co-delivery systems ensure that multiple therapeutics reach the target site simultaneously at optimal ratios, thereby maximizing the therapeutic efficacy of aspirin-based interventions.

Integrating these four strategies, we propose a comprehensive strategy for an aspirin-loaded nanodelivery system tailored to inner ear application, combining ROS-responsive release, LS19-mediated OHC targeting, sustained-release formulation, and potential combination therapy. Beyond these four principal strategies, personalized medicine offers an additional layer of optimization that can further enhance therapeutic efficacy and safety. Leveraging interindividual differences in drug metabolism, genetic profiles (eg., mitochondrial A1555G mutation), and inner ear structure, patient-specific dosing regimens could significantly improve both efficacy and safety.¹⁹¹ Moreover, emerging nanosensor-enabled strategies for real-time salicylate detection allow for tailored aspirin delivery, striking a delicate balance between maximizing therapeutic efficacy and minimizing ototoxic side effects.^{192,193}

Conclusion

In summary, aspirin embodies a paradigm of pharmacological duality in audiology, exhibiting a delicate equilibrium between otoprotection and ototoxicity that is critically governed by dosage, administration route, and individual patient physiology. Low to moderate doses have shown promising potential in preclinical studies to attenuating hearing loss mediated by oxidative stress, inflammation, and cochlear microcirculatory dysfunction. However, high-quality clinical evidence supporting these effects remains limited to specific contexts, and for some indications (eg., ARHL, cisplatin ototoxicity), randomized trials have not demonstrated benefit. Conversely, high-dose application remains fraught with risks of reversible threshold shifts and subtle neural injury. Looking forward, while the future of aspirin in hearing preservation lies not in its conventional use, but potentially in its integration with advanced nanomedicine and personalized therapeutic strategies, it is important to recognize that such approaches remain largely preclinical at this stage. By leveraging targeted nano-delivery systems to achieve site-specific drug accumulation, optimize pharmacokinetics, and minimize systemic adverse effects, aspirin's therapeutic window might be substantially expanded, though significant hurdles related to safety, targeting precision, and clinical translation remain. Future research must prioritize rigorously designed clinical trials with stratified cohorts, long-term follow-up, and multimodal outcome assessments to elucidate dose–response relationships, establish clear clinical guidelines, and validate the efficacy of aspirin-loaded nanoplateforms before any clinical recommendations can be made. Ultimately, the convergence of pharmacology, nanotechnology, and precision medicine may offer new avenues for hearing preservation, but this prospect remains hypothetical pending higher-quality evidence.

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Disclosure

The authors declare no conflicts of interest in this work.

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